

JAI AKKAYARAJKUMAR V J

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Profile

VLSI Graduate student with strong foundation in semiconductor devices, circuit fundamentals, and hardware, software implementation concepts. Skilled in design analysis, trade-off evaluation, and use of industry-standard EDA tools.

Education

M.Tech – VLSI Design	7.9/10
Vellore Institute of Technology (VIT), Vellore	2025–Present
B.E – Electronics and Communication Engineering	8.22/10
Dr. Mahalingam College of Engineering and Technology	2021–2025

Technical Skills

- **Analog / Mixed Signal Design:** CMOS amplifiers, current mirrors, bias circuits, comparators, PLL sub-blocks (PFD, Charge Pump, VCO, Loop Filter), CTLE design, ADC/DAC fundamentals, analog building block analysis, SerDes fundamentals
- **Analog Layout & Physical Verification:** Common-centroid matching, interdigitation, dummy insertion, guard rings, shielding, parasitic-aware routing, antenna fixes, DRC/LVS/PEX sign-off, signal integrity & manufacturability checks
- **Digital Design & ASIC Fundamentals:** Verilog RTL design, FSM implementation (Moore/Mealy), combinational and sequential logic, synthesis flow, STA basics, setup/hold analysis, RTL-to-GDSII flow fundamentals
- **Semiconductor Device Fundamentals:** Short-channel effects, DIBL, channel length modulation, mobility degradation, velocity saturation, body effect, leakage mechanisms, mismatch implications in analog design
- **Physical Design & DFT:** Floorplanning, placement, CTS, routing, timing closure (PD); scan chain insertion, ATPG basics, JTAG boundary scan (DFT)
- **Tools & Languages:** Cadence Virtuoso, Spectre, Innovus, Synopsys tools, MATLAB, Verilog, SystemVerilog, Java, Python, TCL, Linux

Projects

- **AI-Assisted Design and Post-Layout Verification of a 10-Bit SAR ADC(180nm):** Designing, simulating, and laying out a 10-bit SAR ADC in SCL 180 nm CMOS technology using Cadence Virtuoso; developing an XGBoost-based machine learning model to predict initial transistor sizing and analog circuit parameters, reducing design iterations. The design will be validated through DRC, LVS, PEX, and post-layout simulations, targeting 9.4-bit ENOB, 10 MS/s sampling rate, 1.8 V supply voltage, and 1.2 mW power consumption.
- **Single-Stage Active-Inductor CTLE for SerDes (28nm UMC HPC):** Designed and optimized active-inductor based CTLE using Cadence Virtuoso for high-speed channel equalization, analyzed pole-zero behavior for bandwidth enhancement and high-frequency loss compensation, and performed post-layout PEX validation across PVT corners.
- **Analog PLL Design and Layout (180nm SCL PDK):** Implemented PLL sub-blocks including PFD, Charge Pump, VCO, and Loop Filter, performed full-custom layout with matching and shielding techniques, carried out PEX extraction and post-layout simulations, and evaluated loop stability and jitter performance across process corners.
- **CMOS Comparator Layout (180nm):** Designed low-offset CMOS comparator and implemented full-custom layout using common-centroid and symmetry techniques, achieved DRC/LVS clean verification, and analyzed mismatch-induced offset and parasitic effects through post-layout simulations.
- **ADC Behavioral Model for 25-Gbps SerDes:** Developed MATLAB-based ADC behavioral model incorporating jitter, thermal noise, DNL/INL non-idealities, and analyzed ENOB degradation and BER impact under high-speed signaling conditions.
- **RTL-to-GDSII: 8-bit Carry Save Adder:** Implemented complete ASIC design flow from Verilog RTL to GDSII including synthesis, floorplanning, placement, CTS, routing, and static timing analysis, and validated timing closure and physical design constraints using Cadence Innovus.

Certifications & Training

- Insights to the Art of Analog Design — IIT Guwahati
- Analog Layout Design using 180nm SCL PDK
- Analog Circuits and Layout Training — PicoCircuit Technologies
- Custom Analog Layout — VLSI System Design (VSD)